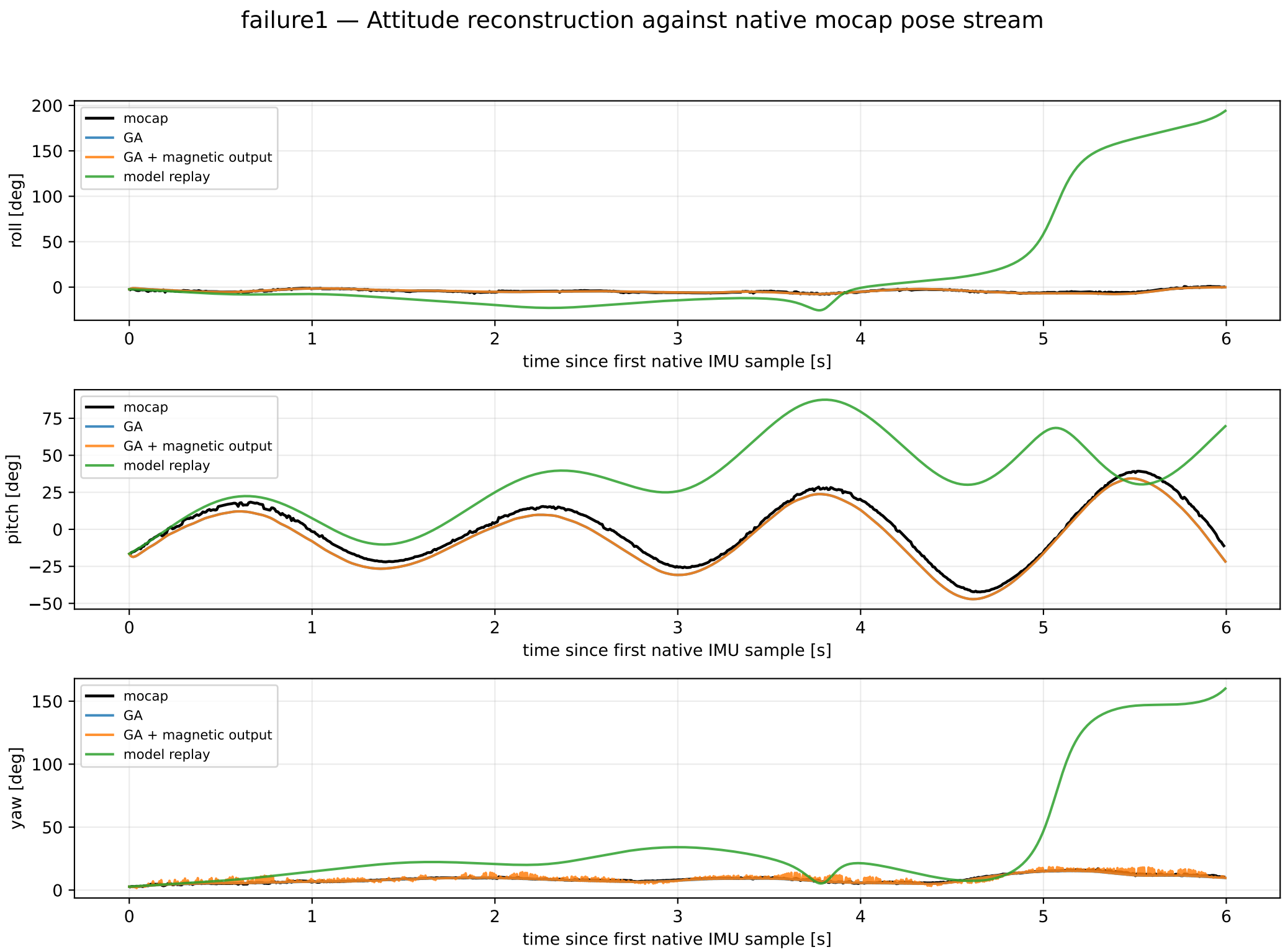
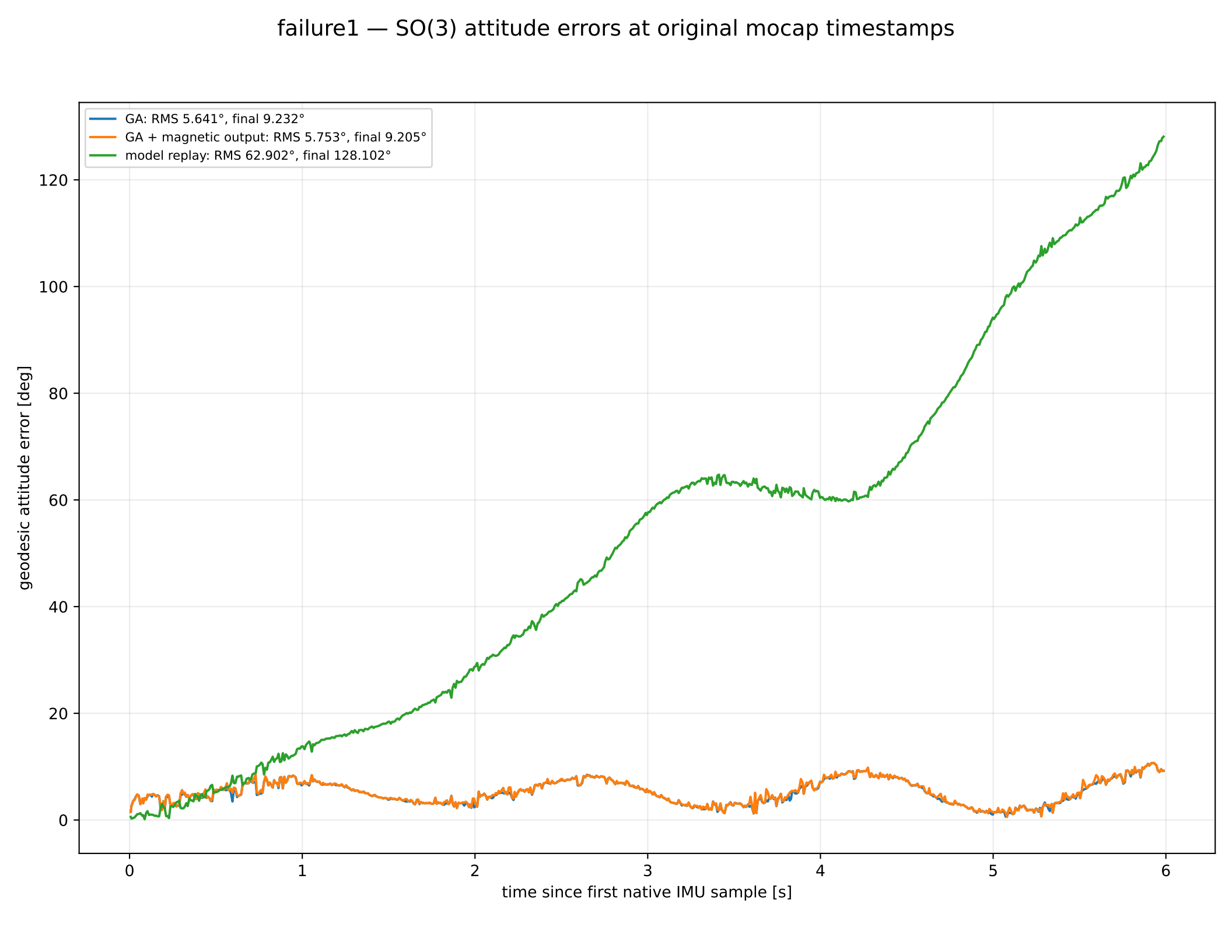
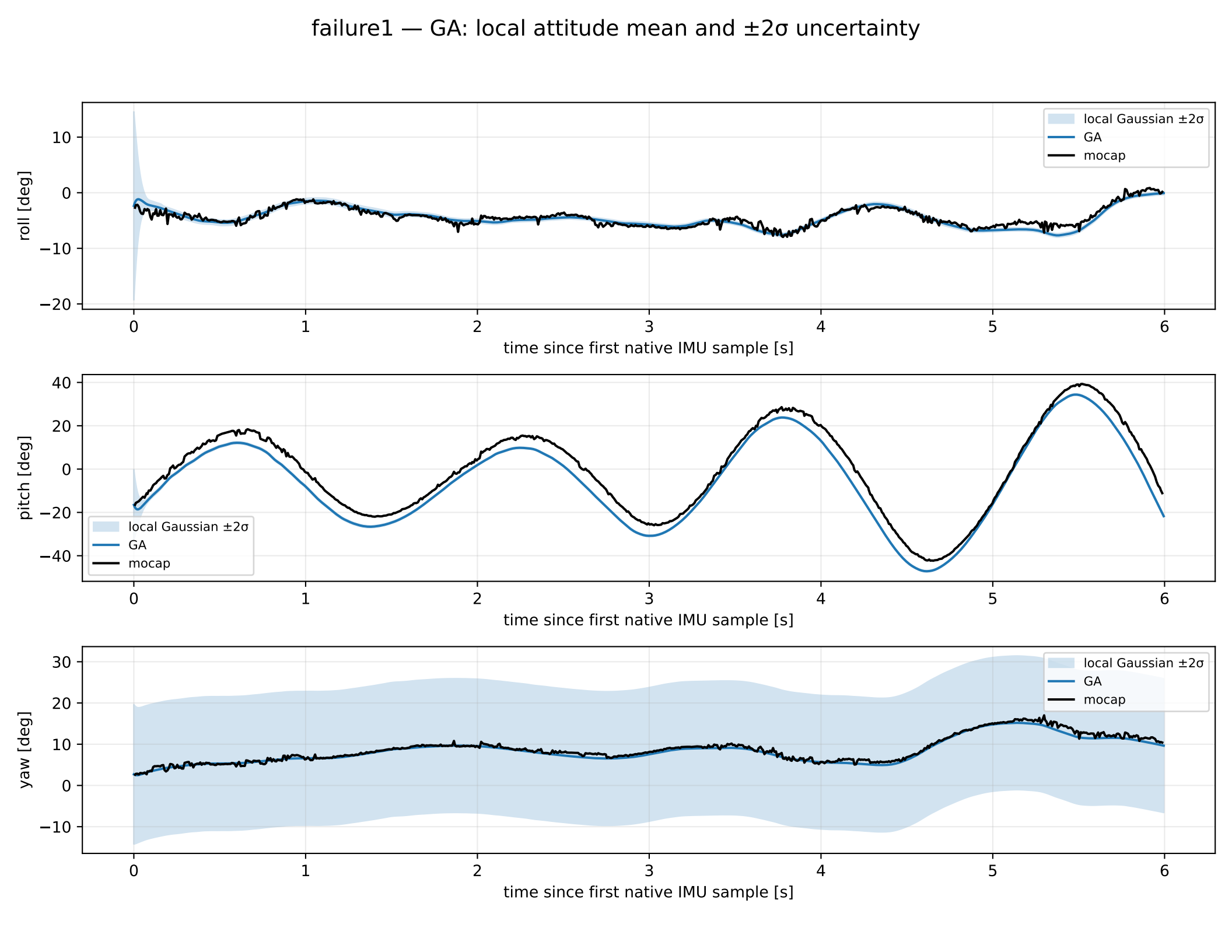
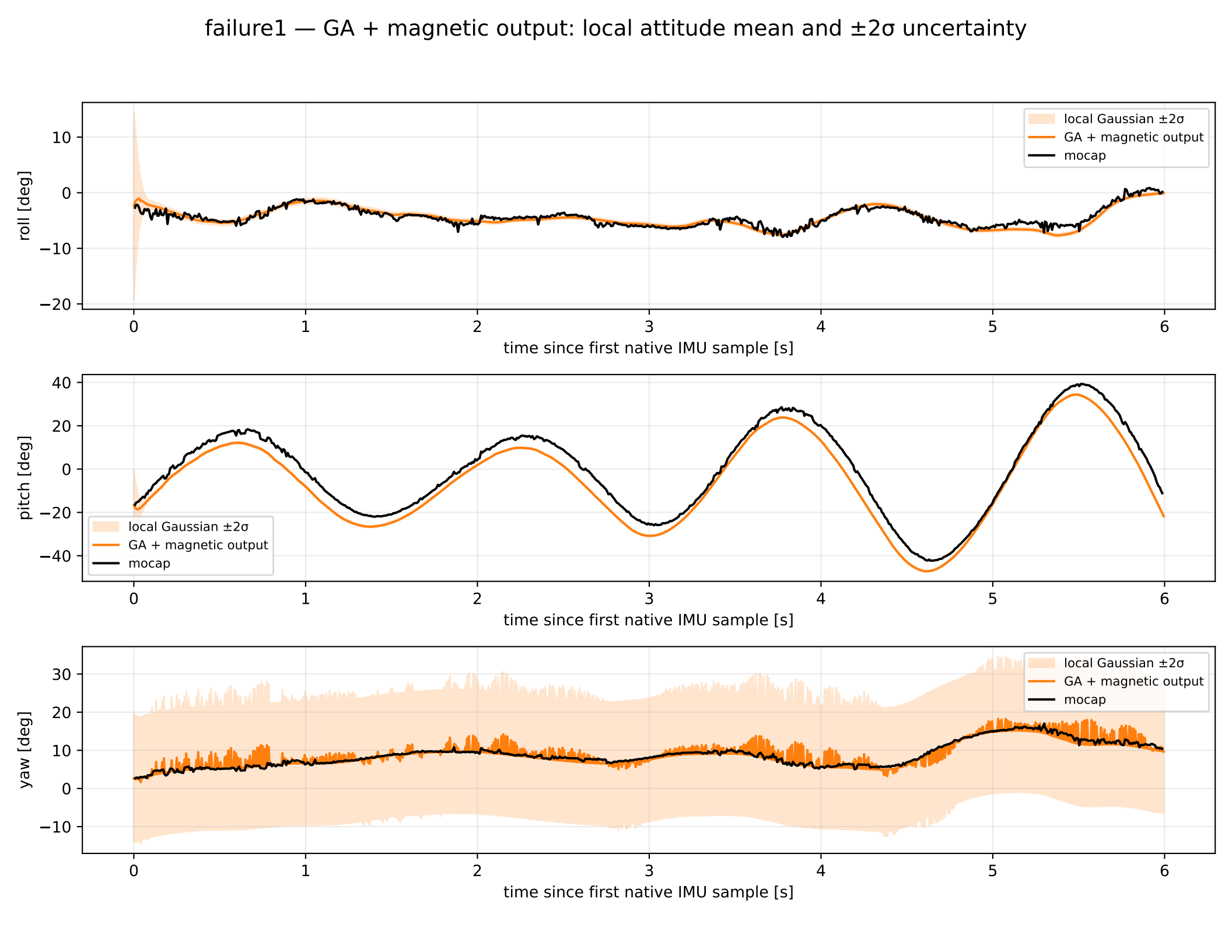
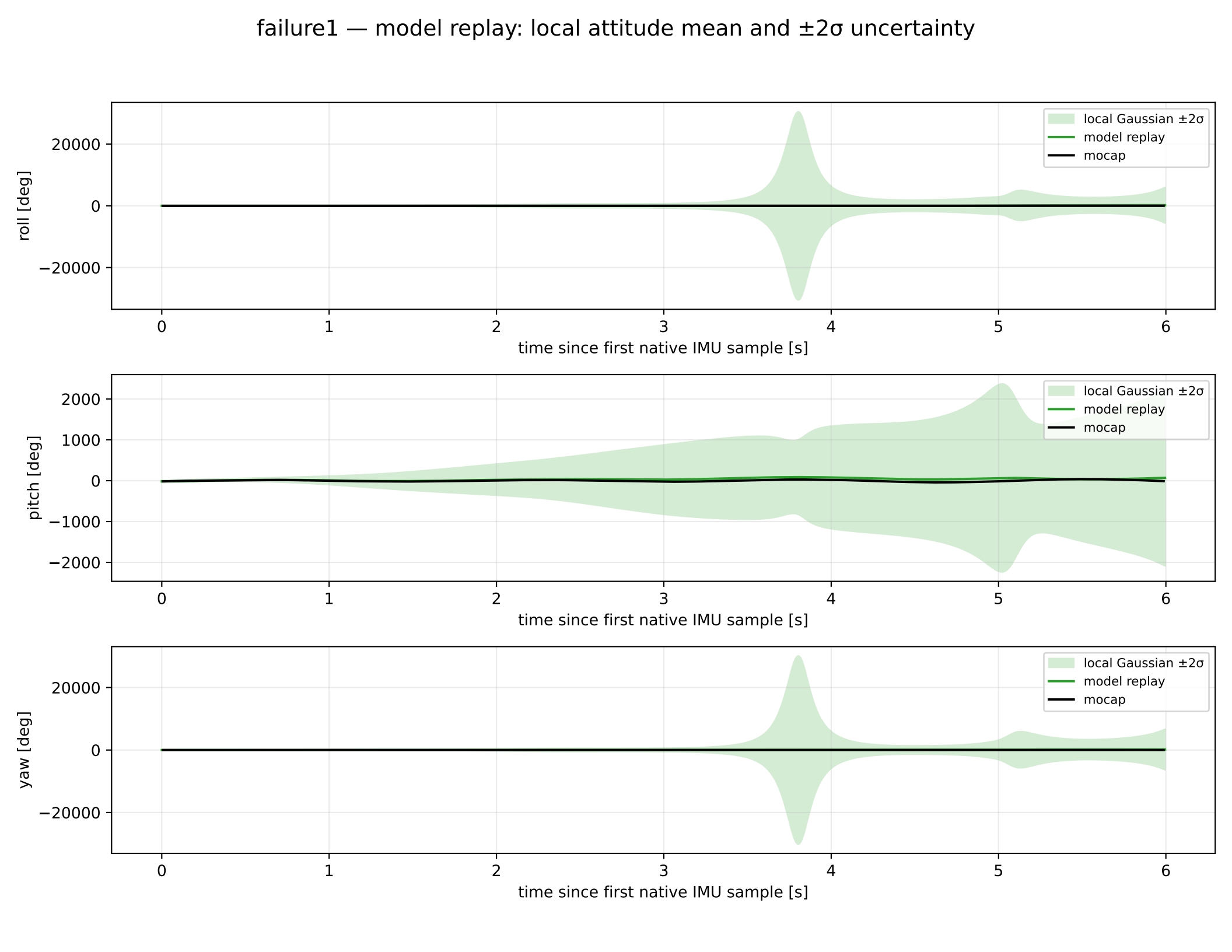

failure1 — Attitude reconstruction against native mocap pose stream
mocap
GA
GA + magnetic output
model replay
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]

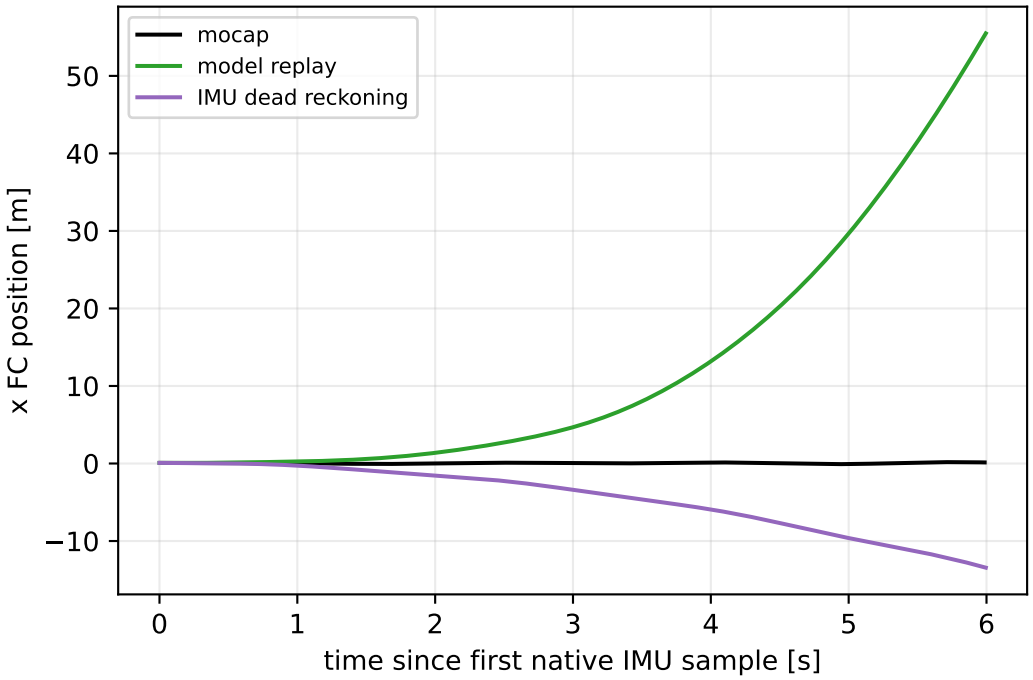

failure1 — SO(3) attitude errors at original mocap timestamps
GA: RMS 5.641°, final 9.232°
GA + magnetic output: RMS 5.753°, final 9.205°
model replay: RMS 62.902°, final 128.102°
geodesic attitude error [deg]
time since first native IMU sample [s]
0
20
40
60
80
100
120
0
1
2
3
4
5
6

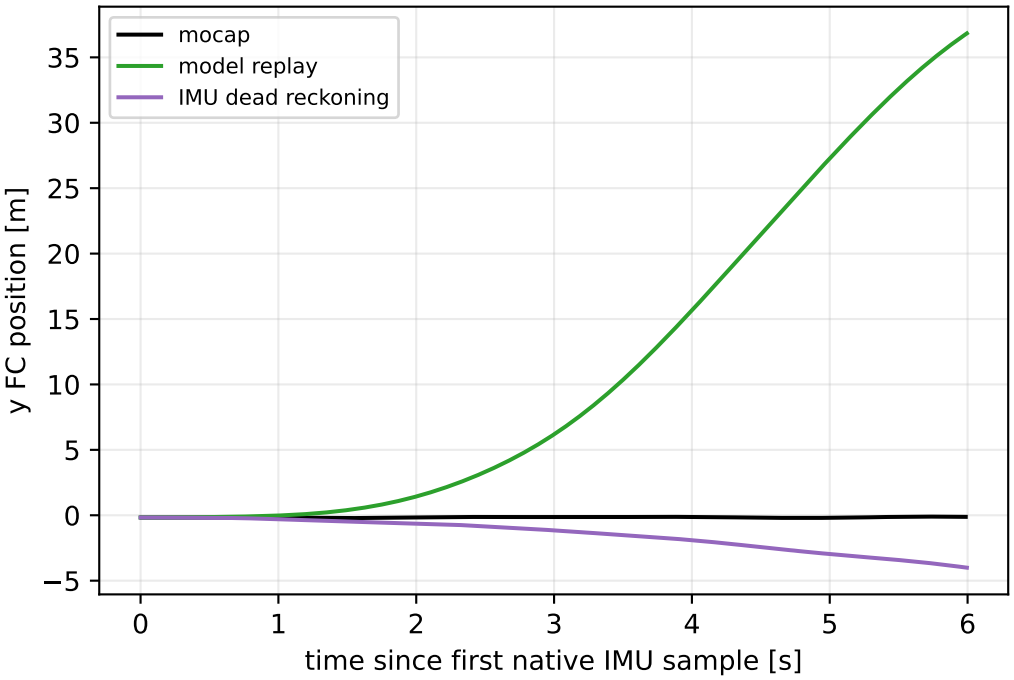

failure1 — GA: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
local Gaussian ±2σ
GA
mocap

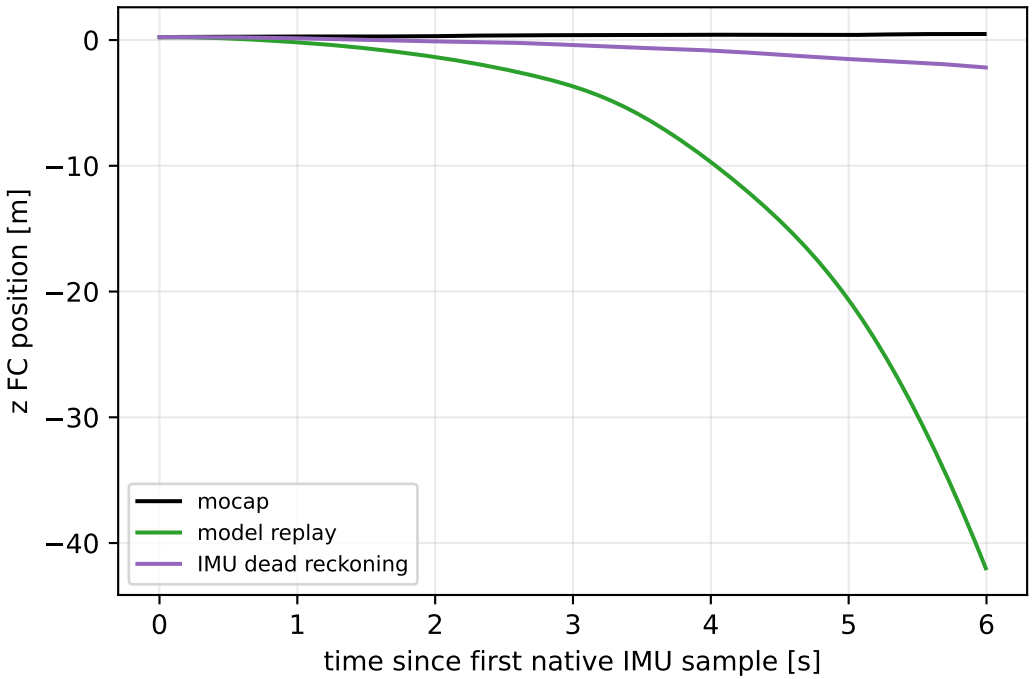

failure1 — GA + magnetic output: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
time since first native IMU sample [s]
local Gaussian ±2σ
GA + magnetic output
mocap

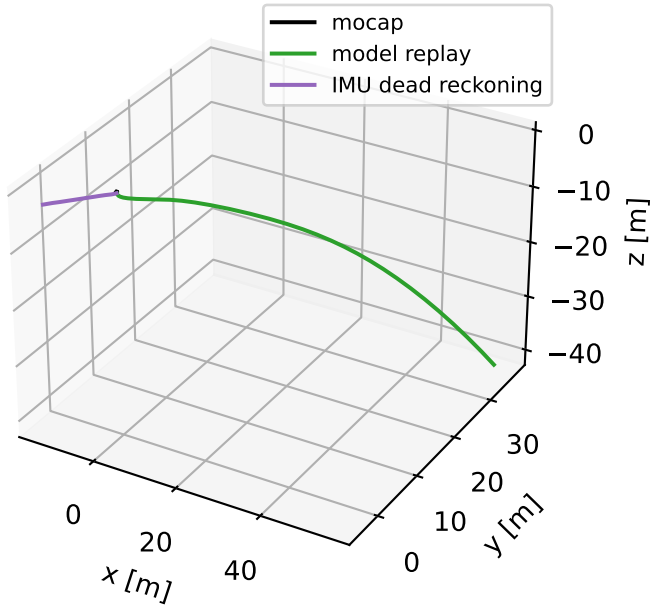

failure1 — model replay: local attitude mean and ±2σ uncertainty
roll [deg]
pitch [deg]
yaw [deg]
local Gaussian ±2σ
model replay
mocap
time since first native IMU sample [s]
20000
0
−20000
2000
1000
−1000
−2000
0
1
2
3
4
5
6

failure1 — Position reconstruction (mean trajectories)



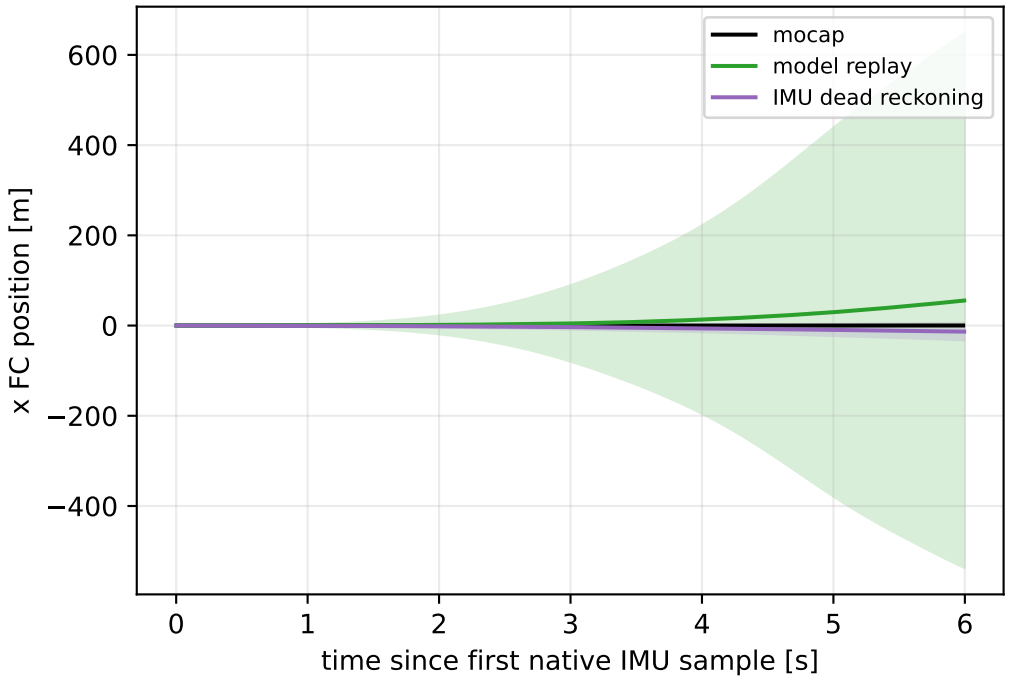


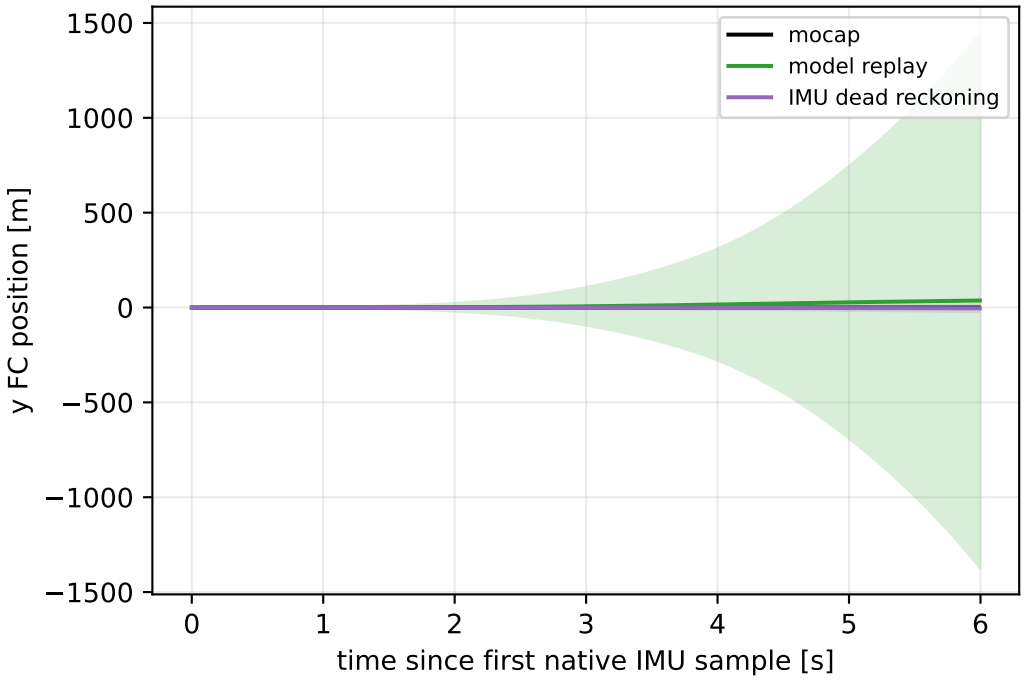


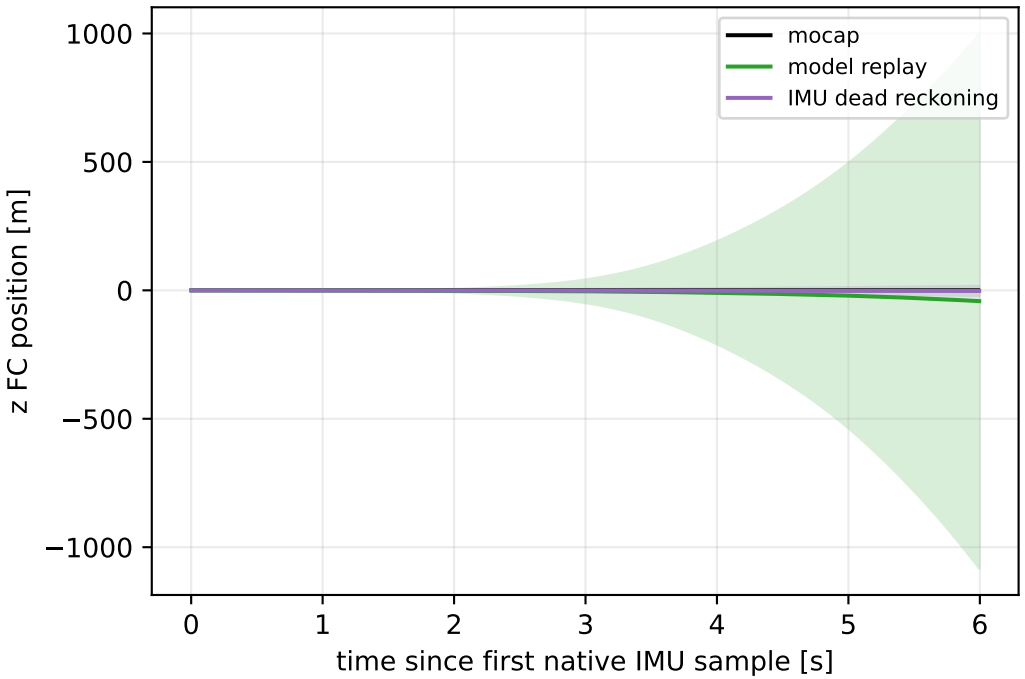


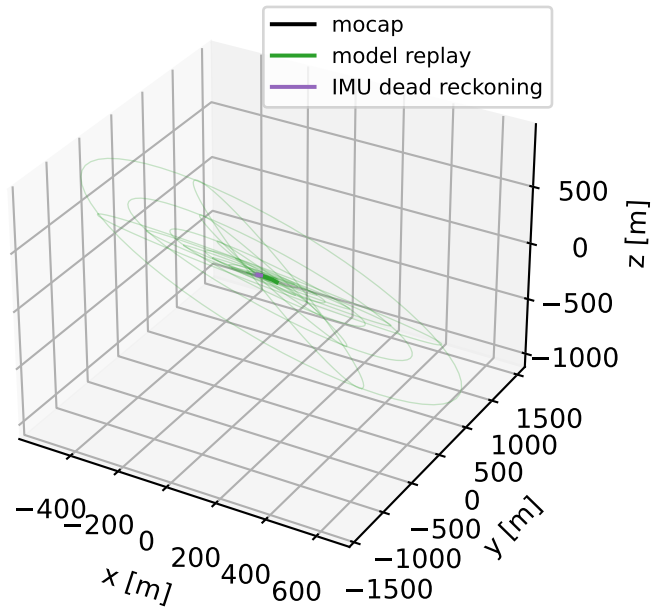
Pure physical replay and IMU integration are not reset to later mocap.

failure1 — Position reconstruction with full propagated ±2σ spread









FC origin; local Gaussian linearization. Ellipsoids: deterministic 2σ factor-SVD wireframes, not samples.